

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **Brucella abortus Stained Suspensions**  
Cat No. : **R30855501**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

Company Oxoid Ltd  
Wade Road  
Basingstoke, Hants, UK  
RG24 8PW  
Tel: +44 (0) 1256 841144

**EU entity/business name**  
Oxoid Deutschland GmbH  
Postfach 10 07 53  
D-46483  
Wesel  
GERMANY  
Tel: + 49 (0) 281 1520  
Fax: 49 (0) 281 1521

E-mail address mbd-sds@thermofisher.com

### 1.4. Emergency telephone number

Chemtrec EU: 001-703-527-3887  
Chemtrec US: (800) 424-9300

For customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

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Based on available data, the classification criteria are not met

## Health hazards

Based on available data, the classification criteria are not met

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements

None required

Signal Word

None

## 2.3. Other hazards

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component    | CAS No  | EC No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                                                              |
|--------------|---------|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Formaldehyde | 50-00-0 | 200-001-8 | <0.1     | Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1 (H317)<br>Carc. 1B (H350)<br>Muta. 2 (H341)<br>STOT SE 3 (H335) |

| Component    | Specific concentration limits (SCL's)                                                                                               | M-Factor | Component notes |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|
| Formaldehyde | Skin Corr. 1B :: C>=25%<br>Eye Irrit. 2 :: 5%<=C<25%<br>Skin Irrit. 2 :: 5%<=C<25%<br>Skin Sens. 1 :: C>=0.2%<br>STOT SE 3 :: C>=5% | -        | -               |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

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|                                           |                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Self-Protection of the First Aider</b> | No special precautions required.                                                                                        |

## **4.2. Most important symptoms and effects, both acute and delayed**

None reasonably foreseeable.

## **4.3. Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

None.

### **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>).

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid contact with skin, eyes or clothing.

### **6.2. Environmental precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

Sweep up and shovel into suitable containers for disposal.

### **6.4. Reference to other sections**

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Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed. Keep at temperatures between 2° and 8 °C.

Technical Rules for Hazardous Substances (TRGS) 510  
Storage Class (LGK) (Germany)

Storage Class/LGK 12

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Third edition. Published 2018. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am arbeitplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

| Component    | European Union                                                                                                           | The United Kingdom                                                                                                      | France                                                | Belgium | Spain                                                                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Formaldehyde | TWA: 0.37 mg/m <sup>3</sup> (8h)<br>TWA: 0.3 ppm (8h)<br>Skin<br>STEL: 0.74 mg/m <sup>3</sup> (8h)<br>STEL: 0.6 ppm (8h) | STEL: 2 ppm 15 min<br>STEL: 2.5 mg/m <sup>3</sup> 15 min<br>TWA: 2 ppm 8 hr<br>TWA: 2.5 mg/m <sup>3</sup> 8 hr<br>Carc. | TWA / VME: 0.5 ppm (8 heures).<br>STEL / VLCT: 1 ppm. |         | STEL / VLA-EC: 0.6 ppm (15 minutos).<br>STEL / VLA-EC: 0.74 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 0.3 ppm (8 horas)<br>TWA / VLA-ED: 0.37 mg/m <sup>3</sup> (8 horas) |

| Component    | Italy                                                                                                                                                                                                                                                         | Germany                                                                                                                                                                                                                                                    | Portugal                                                                                                                                                                                                              | The Netherlands                                                              | Finland                                                                                                                                          |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Formaldehyde | TWA: 0.37 mg/m <sup>3</sup> 8 ore.<br>Media Ponderata nel Tempo<br>TWA: 0.3 ppm 8 ore.<br>Media Ponderata nel Tempo<br>TWA: 0.62 mg/m <sup>3</sup> 8 ore.<br>Media Ponderata nel Tempo for the health care, funeral and embalming sectors until July 11, 2024 | TWA: 0.3 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 0.37 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 0.3 ppm (8 Stunden). MAK no irritation should occur during mixed exposure<br>TWA: 0.37 mg/m <sup>3</sup> (8 Stunden). MAK no | STEL: 0.6 ppm 15 minutos<br>STEL: 0.74 mg/m <sup>3</sup> 15 minutos<br>Ceiling: 0.3 ppm<br>TWA: 0.3 ppm 8 horas<br>TWA: 0.37 mg/m <sup>3</sup> 8 horas<br>TWA: 0.62 mg/m <sup>3</sup> 8 horas<br>TWA: 0.5 ppm 8 horas | STEL: 0.5 mg/m <sup>3</sup> 15 minuten<br>TWA: 0.15 mg/m <sup>3</sup> 8 uren | TWA: 0.3 ppm 8 tunteina<br>TWA: 0.37 mg/m <sup>3</sup> 8 tunteina<br>STEL: 0.6 ppm 15 minuutteina<br>STEL: 0.74 mg/m <sup>3</sup> 15 minuutteina |

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|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--|--|--|
|  | TWA: 0.5 ppm 8 ore.<br>Media Ponderata nel<br>Tempo for the health<br>care, funeral and<br>embalming sectors until<br>July 11, 2024<br>STEL: 0.74 mg/m <sup>3</sup> 15<br>minuti. Breve termine<br>STEL: 0.6 mg/m <sup>3</sup> 15<br>minuti. Breve termine<br>Pelle | irritation should occur<br>during mixed exposure<br>Höhepunkt: 0.6 ppm<br>Höhepunkt: 0.74 mg/m <sup>3</sup> |  |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--|--|--|

| Component    | Austria                                                                                                                                                            | Denmark                                            | Switzerland                                                                                                                                        | Poland                                                                                    | Norway                                                                                                         |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Formaldehyde | MAK-KZGW: 0.6 ppm<br>15 Minuten<br>MAK-KZGW: 0.74<br>mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 0.3 ppm 8<br>Stunden<br>MAK-TMW: 0.37 mg/m <sup>3</sup><br>8 Stunden | Ceiling: 0.3 ppm<br>Ceiling: 0.4 mg/m <sup>3</sup> | STEL: 0.6 ppm 15<br>Minuten<br>STEL: 0.74 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 0.3 ppm 8<br>Stunden<br>TWA: 0.37 mg/m <sup>3</sup> 8<br>Stunden | STEL: 0.74 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 0.37 mg/m <sup>3</sup> 8<br>godzinach | TWA: 0.5 ppm 8 timer<br>TWA: 0.6 mg/m <sup>3</sup> 8 timer<br>Ceiling: 1 ppm<br>Ceiling: 1.2 mg/m <sup>3</sup> |

| Component    | Bulgaria                                                   | Croatia                                                                                                                                                                                                                                                                                                                                                                                                               | Ireland                                                                                                                                                                                                                                                                                                                                                                                          | Cyprus | Czech Republic                                                                                                     |
|--------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------|
| Formaldehyde | TWA: 1.0 mg/m <sup>3</sup><br>STEL : 2.0 mg/m <sup>3</sup> | TWA-GVI: 0.3 ppm 8<br>satima.<br>TWA-GVI: 0.37 mg/m <sup>3</sup> 8<br>satima.<br>TWA-GVI: 0.5 ppm 8<br>satima. for health,<br>funeral and embalming<br>sector applies until July<br>11, 2024<br>TWA-GVI: 0.62 mg/m <sup>3</sup> 8<br>satima. for health,<br>funeral and embalming<br>sector applies until July<br>11, 2024<br>STEL-KGVI: 0.6 ppm 15<br>minutama.<br>STEL-KGVI: 0.74 mg/m <sup>3</sup><br>15 minutama. | TWA: 0.3 ppm 8 hr.<br>TWA: 0.5 ppm 8 hr. for<br>the healthcare, funeral<br>and embalming sectors<br>until July 11, 2024<br>TWA: 0.37 mg/m <sup>3</sup> 8 hr.<br>TWA: 0.62 mg/m <sup>3</sup> 8 hr.<br>for the healthcare,<br>funeral and embalming<br>sectors until July 11,<br>2024<br>STEL: 0.6 ppm 15 min<br>STEL: 0.738 mg/m <sup>3</sup> 15<br>min<br>STEL: 0.62 mg/m <sup>3</sup> 15<br>min |        | TWA: 0.5 mg/m <sup>3</sup> 8<br>hodinách.<br>Potential for cutaneous<br>absorption<br>Ceiling: 1 mg/m <sup>3</sup> |

| Component    | Estonia                                                                                                                                                  | Gibraltar | Greece                                                                                       | Hungary                                                                                                                                     | Iceland                                                                                                                                               |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Formaldehyde | TWA: 0.5 ppm 8<br>tundides.<br>TWA: 0.6 mg/m <sup>3</sup> 8<br>tundides.<br>STEL: 1 ppm 15<br>minutites.<br>STEL: 1.2 mg/m <sup>3</sup> 15<br>minutites. |           | STEL: 0.6 ppm<br>STEL: 0.74 mg/m <sup>3</sup><br>TWA: 0.3 ppm<br>TWA: 0.37 mg/m <sup>3</sup> | STEL: 0.6 mg/m <sup>3</sup> 15<br>percekben. CK<br>TWA: 0.6 mg/m <sup>3</sup> 8<br>óraban. AK<br>lehetséges borön<br>keresztüli felszívódás | STEL: 0.6 ppm<br>STEL: 0.74 mg/m <sup>3</sup><br>TWA: 0.3 ppm 8<br>klukkustundum.<br>TWA: 0.37 mg/m <sup>3</sup> 8<br>klukkustundum.<br>Skin notation |

| Component    | Latvia                     | Lithuania                                                                                                                                                                                                                                                                          | Luxembourg | Malta | Romania                                                                                                                 |
|--------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|-------------------------------------------------------------------------------------------------------------------------|
| Formaldehyde | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.3 ppm IPRD<br>TWA: 0.37 mg/m <sup>3</sup> IPRD<br>TWA: 0.62 mg/m <sup>3</sup> IPRD<br>for healthcare, funeral,<br>and embalming<br>industries<br>TWA: 0.5 ppm IPRD for<br>healthcare, funeral, and<br>embalming industries<br>STEL: 0.74 mg/m <sup>3</sup><br>STEL: 0.6 ppm |            |       | TWA: 1 ppm 8 ore<br>TWA: 1.2 mg/m <sup>3</sup> 8 ore<br>STEL: 2 ppm 15 minute<br>STEL: 3 mg/m <sup>3</sup> 15<br>minute |

| Component    | Russia                                      | Slovak Republic                                 | Slovenia                                                 | Sweden                              | Turkey |
|--------------|---------------------------------------------|-------------------------------------------------|----------------------------------------------------------|-------------------------------------|--------|
| Formaldehyde | Skin notation<br>MAC: 0.5 mg/m <sup>3</sup> | Ceiling: 0.74 mg/m <sup>3</sup><br>TWA: 0.3 ppm | TWA: 0.62 mg/m <sup>3</sup> 8<br>urah applies for health | Binding STEL: 0.6 ppm<br>15 minuter |        |

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|  |  |                             |                                                                                                                                                                                                                                                                                                             |                                                                                                                                   |  |
|--|--|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--|
|  |  | TWA: 0.37 mg/m <sup>3</sup> | care, funeral and embalming activities until July 11, 2024<br>TWA: 0.5 ppm 8 urah<br>applies for health care, funeral and embalming activities until July 11, 2024<br>TWA: 0.37 mg/m <sup>3</sup> 8 urah<br>TWA: 0.3 ppm 8 urah Koža<br>STEL: 0.6 ppm 15 minutah<br>STEL: 0.74 mg/m <sup>3</sup> 15 minutah | Binding STEL: 0.74 mg/m <sup>3</sup> 15 minuter<br>TLV: 0.3 ppm 8 timmar. NGV<br>TLV: 0.37 mg/m <sup>3</sup> 8 timmar. NGV<br>Hud |  |
|--|--|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--|

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                     | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Formaldehyde 50-00-0 ( <0.1 ) |                              |                                 | DNEL = 37µg/cm <sup>2</sup>    | DNEL = 240mg/kg bw/day            |

| Component                     | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Formaldehyde 50-00-0 ( <0.1 ) | DNEL = 0.75mg/m <sup>3</sup>     |                                     | DNEL = 0.375mg/m <sup>3</sup>      | DNEL = 9mg/m <sup>3</sup>             |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                     | Fresh water     | Fresh water sediment        | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)      |
|-------------------------------|-----------------|-----------------------------|--------------------|------------------------------------|-------------------------|
| Formaldehyde 50-00-0 ( <0.1 ) | PNEC = 0.44mg/L | PNEC = 2.3mg/kg sediment dw | PNEC = 4.44mg/L    | PNEC = 0.19mg/L                    | PNEC = 0.2mg/kg soil dw |

| Component                     | Marine water    | Marine water sediment       | Marine water Intermittent | Food chain | Air |
|-------------------------------|-----------------|-----------------------------|---------------------------|------------|-----|
| Formaldehyde 50-00-0 ( <0.1 ) | PNEC = 0.44mg/L | PNEC = 2.3mg/kg sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

None under normal use conditions.

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## Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material    | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

**Skin and body protection** Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particle filter

**Small scale/Laboratory use** Maintain adequate ventilation

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Blue                     |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>Melting Point/Range</b>                     | Not applicable           |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | Not applicable           |                                          |
| <b>Flammability (liquid)</b>                   | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | Not applicable           | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | Not applicable           |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>pH</b>                                      | 7.3 - 7.5                |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Soluble                  |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Formaldehyde                                   | -0.35                    |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | No data available        |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

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## 9.2. Other information

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under recommended storage conditions.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization

Hazardous polymerization does not occur.

#### Hazardous Reactions

None under normal processing.

### 10.4. Conditions to avoid

Exposure to moisture.

### 10.5. Incompatible materials

None known.

### 10.6. Hazardous decomposition products

Nitrogen oxides (NOx).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

Product does not present an acute toxicity hazard based on known or supplied information

#### (a) acute toxicity;

##### Oral

Based on available data, the classification criteria are not met

##### Dermal

Based on available data, the classification criteria are not met

##### Inhalation

Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component    | LD50 Oral       | LD50 Dermal               | LC50 Inhalation      |
|--------------|-----------------|---------------------------|----------------------|
| Formaldehyde | 500 mg/kg (Rat) | LD50 = 270 mg/kg (Rabbit) | 0.578 mg/L (Rat) 4 h |

#### (b) skin corrosion/irritation;

No data available

#### (c) serious eye damage/irritation;

No data available

#### (d) respiratory or skin sensitization;

##### Respiratory

No data available

##### Skin

No data available

| Component                        | Test method                                  | Test species      | Study result                |
|----------------------------------|----------------------------------------------|-------------------|-----------------------------|
| Formaldehyde<br>50-00-0 ( <0.1 ) | Skin sensitization<br>Test method Patch Test | Man<br>guinea pig | Sensitizer<br>Sensitization |



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|  |                                       |  |  |
|--|---------------------------------------|--|--|
|  | Respiratory sensitization<br>in vitro |  |  |
|--|---------------------------------------|--|--|

No information available

(e) germ cell mutagenicity; No data available

None known

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component    | EU           | UK    | Germany | IARC    |
|--------------|--------------|-------|---------|---------|
| Formaldehyde | Carc Cat. 1B | Cat 3 |         | Group 1 |

(g) reproductive toxicity;  
Reproductive Effects  
Developmental Effects

No data available  
None known.  
None known.

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects

| Component    | Freshwater Fish                       | Water Flea                              | Freshwater Algae |
|--------------|---------------------------------------|-----------------------------------------|------------------|
| Formaldehyde | Leuciscus idus: LC50 = 15 mg/L<br>96h | EC50 = 20 mg/L 96h<br>EC50 = 2 mg/L 48h |                  |

### 12.2. Persistence and degradability

**Persistence** Soluble in water, Persistence is unlikely, based on information available.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component    | log Pow | Bioconcentration factor (BCF) |
|--------------|---------|-------------------------------|
| Formaldehyde | -0.35   | No data available             |

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|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>12.4. Mobility in soil</u></b>                                                                                 | The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils |
| <b><u>12.5. Results of PBT and vPvB assessment</u></b>                                                               | No data available for assessment.                                                                                                                            |
| <b><u>12.6. Endocrine disrupting properties</u></b><br><b>Endocrine Disruptor Information</b>                        | This product does not contain any known or suspected endocrine disruptors                                                                                    |
| <b><u>12.7. Other adverse effects</u></b><br><b>Persistent Organic Pollutant</b><br><b>Ozone Depletion Potential</b> | This product does not contain any known or suspected substance<br>This product does not contain any known or suspected substance                             |

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

|                                            |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.                                                                                |
| <b>Contaminated Packaging</b>              | Dispose of in accordance with local regulations.                                                                                                                                                                                                                                                           |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used.                                                                                                                                                                                                        |
| <b>Switzerland - Waste Ordinance</b>       | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**ADR** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**IATA** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**

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## 14.3. Transport hazard class(es)

## 14.4. Packing group

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component    | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Formaldehyde | 50-00-0 | 200-001-8 | -      | -   | X     | X    | KE-17074 | X    | X    |

| Component    | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------|---------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Formaldehyde | 50-00-0 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

| Component    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                         | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Formaldehyde | -                                                                   | Use restricted. See item 72. (see link for restriction details)<br>Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

| Component    | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Formaldehyde | 50-00-0 | 5 tonne                                                                                   | 50 tonne                                                                                 |

#### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work  
Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work.

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

#### National Regulations

OXDR30855501

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UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** Water endangering class = non-hazardous to waters (self classification)

| Component    | Germany - Water Classification (VwVwS) | Germany - TA-Luft Class                              |
|--------------|----------------------------------------|------------------------------------------------------|
| Formaldehyde | WGK 3                                  | Class I : 20 mg/m <sup>3</sup> (Massenkonzentration) |

| Component    | France - INRS (Tables of occupational diseases)      |
|--------------|------------------------------------------------------|
| Formaldehyde | Tableaux des maladies professionnelles (TMP) - RG 43 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                      | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|--------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Formaldehyde<br>50-00-0 (<0.1) |                                                                                                                | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H301 - Toxic if swallowed  
H311 - Toxic in contact with skin  
H331 - Toxic if inhaled  
H314 - Causes severe skin burns and eye damage  
H318 - Causes serious eye damage  
H317 - May cause an allergic skin reaction  
H351 - Suspected of causing cancer  
H335 - May cause respiratory irritation  
H341 - Suspected of causing genetic defects  
H350 - May cause cancer

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

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**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Creation Date** 04-Jul-2011

**Revision Date** 10-Dec-2021

**Revision Summary** Update to CLP Format.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**